

The Price of Thought: A Multilingual Analysis of Reasoning, Performance, and Cost of Negotiation in Large Language Models

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Abstract

This document is a supplements to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers. This document is a supplements to the general instructions for *ACL authors. It contains instructions for using the L^AT_EX style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

1 Introduction

Negotiation is a key aspect of human social and economic behaviour that poses significant challenges for AI systems. The problem is complex where agents must produce coherent language while performing strategic reasoning, modelling the opponents' preferences and goals, and managing the tension between cooperation and competition to maximize own outcomes. With LLMs increasingly being tested as autonomous agents in domains ranging from e-commerce to resource distribution with new protocols for agents being developed, e.g. Agents to Payments (AP2) protocol (Google Cloud, 2025). It has become essential to assess the limitations and capabilities of language agents, or vanilla LLMs for negotiation. Negotiation tasks provide a strong test-bed for measuring agent competence,

offering interactive, multi-turn scenarios with objective measures that go beyond static evaluation.

Studies show that LLMs often deviate from optimal play with unreliable performance, where even top-performing systems can lose to weaker opponents and fail in cooperative scenarios (Davidson et al., 2024; Hua et al., 2024; Bianchi et al., 2024; Kwon et al., 2024). These models can acquire deceptive tactics, e.g., expressing false interest in low-value items for later concessions (Lewis et al., 2017), express desperation to improve outcomes (Bianchi et al., 2024), perform much worse as buyers than sellers (Xia et al., 2024), or even take economic risks that lead to budget violations and overpayment (Zhu et al., 2025).

Despite this growing body of research, two critical gaps remain unexplored. First, no previous work has systematically investigated how Chain-of-Thought prompting and reasoning capabilities influence negotiation abilities in terms both performance and computational cost, a fundamental oversight given that reasoning is central to strategic decision-making. Second, the field has been constrained by monolingual English-only analyses, leaving multilingual negotiation capabilities completely unexplored. To address these significant limitations, we conduct the first comprehensive study examining reasoning effects on negotiation tasks across three languages: English, German, and Italian. We implement three distinct dialogue games requiring bargaining skills, collaboration, and strategic preference management (von Neumann and Morgenstern, 1944; Nash, 1953; Fisher et al., 2011), testing both commercial and open-weight LLMs in a self-play mode where both sides of the games are played by themselves. We analyse the impact of systematically enabling and disabling reasoning mode of evaluated models.

Our investigation targets three fundamental research questions:

What is the computational and performance

trade-off between reasoning overhead and negotiation effectiveness across different tasks and languages?

To what extent do models maintain language consistency in their reasoning processes when performing multilingual negotiation tasks?

Do models demonstrate genuine strategic adaptation over multiple turns, or merely surface-level pattern matching that creates an illusion of learning?

2 Related Work

Game theory and negotiation research has deep foundations in the analysis of strategic interactions between rational agents, with seminal contributions from (Nash, 1950) on bargaining solutions, (Rubinstein, 1982) on sequential bargaining, and (Raiffa, 1985) on negotiation analysis. The convergence of Large Language Models (LLMs) and game-theoretic frameworks represents a significant development in the evaluation and understanding of strategic capabilities in AI systems (Sun et al., 2025). Negotiation games serve as demanding test environments for LLMs, requiring integration of linguistic competence, strategic planning, and social reasoning capabilities.

Benchmarking strategic capabilities in LLMs: systematic evaluation of LLM negotiation performance reveals substantial variance in strategic behaviour, with models frequently departing from optimal play and exhibiting asymmetric performance patterns (Davidson et al., 2024; Hua et al., 2024; Akata et al., 2023) or when some powerful models underperform against weaker opponents and encounter difficulties in cooperative bargaining scenarios. This has motivated the creation of comprehensive evaluation frameworks and experimental platforms designed to assess agent (initialised as LLMs) behaviour across various negotiation contexts, including resource allocation and pricing tasks (Bianchi et al., 2024; Xia et al., 2024). Empirical studies reveal that LLMs can display irrational tactics, e.g. deceptive strategies such as expressing false interest in low-value items to create bargaining leverage in subsequent exchanges (Lewis et al., 2017). LLMs also exhibit susceptibility to cognitive biases including anchoring effects and responsiveness to social manipulation tactics, where expressions of desperation or aggressive language can substantially affect negotiation outcomes (Bianchi et al., 2024). However, these behaviours present

practical risks, including budget constraint violations and acceptance of economically disadvantageous agreements (Zhu et al., 2025). A consistent finding across studies is the asymmetric difficulty in buyer versus seller roles, with agents systematically underperforming as buyers (Xia et al., 2024).

Improving strategic reasoning and decision-making: Given the inconsistent strategic reasoning observed in current LLMs (Wong et al., 2025), research efforts focus on enhancing decision-making through structured approaches such as integration of game-theoretic solvers with LLM dialogue (Gemp et al., 2024), structured reasoning workflows based on Dominant Strategy Search and Backward Induction (Hua et al., 2024), or even learn by interacting based with reinforcement learning (Cao et al., 2018). From prompting perspective, Chain-of-Thought reasoning emerges as a critical factor in agent performance (Gandhi et al., 2023), contributing to more consistent outcomes in integrative negotiation settings (Vaccaro et al., 2025). Hybrid architectural approaches have been developed to constrain agent behaviour, such as employing deterministic offer generation modules to control pricing decisions while utilising LLMs for natural language dialogue production, resulting in substantial improvements in deal completion rates and profit margins (Xia et al., 2024).

3 Evaluating Negotiation Abilities with Dialogue Games

In this section, we provide details on how we implement three dialogue games to test negotiation abilities in LLMs. Each game has its unique properties on what specifically it measures: bargaining, collaboration, grounding, or personal preference maximization. We define a dialogue game as a structured communication between two agents (LLMs) according to a given communication protocol towards the defined goal. The goals in games are defined as the expected outcome, e.g. to make a deal, that agents attempt to maximize. The communication protocols dictates how messages are expected to be formatted. A negotiation task unfolds by agents starting to communicate with one another for the given goal and the conversation continues until it reaches one of the defined stopping criteria: 1) goal state is reached, 2) maximum turn limit is reached, 3) agents violate the communication protocol and the game-play is aborted. The orchestration of this message passing between agents,

validation of communication protocols, checking whether goal states are reached or not are done by Game Master, which is a scripted entity in the dialogue.

We use the *clembench framework* (Chalamalasetti et al., 2023) to implement the dialogue games. The framework already includes all necessary components regarding integration of commercial and open-weight LLMs, goal state definitions and scoring, the definitions of agents and the Game Master. The dialogue games are run in a self-play mode where both participants of the game are LLMs playing against or with one another.

Next, we provide details about each specific game.

3.1 Deal or No Deal

The Deal or No Deal (DoND) game is a two-player game designed to simulate a multi-issue bargaining scenario. DoND has previously been proposed in (Lewis et al., 2017) to training an AI system for negotiation. More recently, Liao et al. (2024) used the same DoND game and proposed three different variations, two of which are used in this work. Figure 1 shows an example of a successfully played episode of the game. Two players must communicate to reach a mutually beneficial agreement on the division of a set of items, each holding different values for each player. The game focuses on evaluating negotiation skills with a focus on the ability to express and understand preferences, and to compromise. We provide all prompts and other details in Appendix B.

Game Mechanics Two agents negotiate over a shared set of items, each having different private value functions for item types. Players exchange free-form messages and then simultaneously make secret proposals using specific syntax. If proposals are compatible (enough items for both), each player receives points based on their requested items’ value according to their private function. If proposals conflict, both receive zero points. The game master strictly enforces rules without re-prompting, making rule adherence critical for successful play.

Game Instances We generate instances for the game programmatically by randomly sampling 100 common nouns. Each instance was generated with a maximum turn limit of 5 and each game instance features a random set of between 3 and 5 different item types. We have two different sets of experiments. In the **semi-competitive** experiment, play-

#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... Available items: 1 book, 2 hats, 2 balls. Player A value function: book: 0, hat: 1, ball: 4.	
2		[GM(B)] <GAME DESCRIPTION> Player B value function: book: 6, hat: 1, ball: 1.	
3	[A(B)] Hi! Here are my preferences: I value balls the most, then hats, and I don't value the book at all. How about you? What do you value most?		
4		[A(B)] Thanks for sharing! I value the book most, while hats and balls are equally less important to me. Would you take both balls and one hat, and I'll take the book and the other hat? That way, we both get what we value most.	
5	[A(B)] That sounds fair! With this split, I'd get both balls and one hat, while you'd get the book and one hat, so we each get what we value most. Ready to submit this as my proposal unless you'd like to tweak anything.		
6		[A(B)] I agree with the split. I'll take the book and one hat, you take both balls and the other hat.	
7	[A(GM)] [Proposal: 1 hat, 2 balls]		
8		[GM(B)] The other player has submitted a proposal. You must now submit your proposal.	
9		[GM(B)] [Proposal: 1 book, 1 hat]	

Figure 1: An example of a *Deal or No Deal* game episode. Note that the initial prompts given by the game master have been truncated for brevity. The game ends in a successful Pareto optimal agreement.

ers are asked to maximize their own score. This is the standard negotiation setting, and it requires players to balance cooperation, to avoid zero-point conflicts, with competition, to maximize their individual share. For the **cooperative** experiment, players are asked to maximize the sum of the scores of both players. This transforms the game into a purely cooperative task, in which two rational agents should always achieve the maximum score, assuming they are able to communicate and reason about the preferences perfectly. Each experiment includes 20 instances, with **total 40 instances**.

3.2 Clean Up

Clean Up is a two-player game focused on cooperative strategy development and object rearrangement. Object rearrangement has been a long-standing problem that requires modelling the given situation, the action space, and spatial reasoning (Batra et al., 2020; Zeng et al., 2024; Khan et al., 2025). In this game, the players need to both negotiate their common goal state and a strategy of reaching it (Jeknic et al., 2024). A sample episode is shown in Figure 2. Each player can only access its own grid and need to negotiate which items to rearrange so that the grids become identical at the end. We provide all prompts and other details in Appendix C.

Game Mechanics Both players are presented with 7×7 ASCII grids, with a number of randomly distributed objects in the form of capital letters placed on them. In each turn, a player can either send a message to their counterpart or move an object on their grid. The message and move text require certain formatting rules to follow. If a player does not follow the format, tries to move an object to a non-empty space or outside the grid bounds, or tries to move an object that doesn't exist, they receive a penalty and are re-prompted with information on the nature of their mistake. The game ends if (1) both players agree to end it, (2) round limit is exceeded, or (3) penalty limit is exceeded.

#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... 1234567 <pre> Coo 1 oIo 2 ooo 3 ooo 4 ooo 5 oWT 6 ooh 7 </pre> Please send a message to the other player to start the game!	
2	[A(GM)] SAY: Hi! Let's coordinate our moves. [...]	[GM(B)] <GAME DESCRIPTION> ... 1234567 <pre> ooo 1 oWo 2 ooo 3 ooh 4 T 5 oIo 6 ooo 7 </pre> The other player started the game by sending this message: <MESSAGE> What is your first command?	
3			
4		[GM(B)] MOVE: C, (1,1) [A(GM)] Your message has been relayed to the other player.	
5		<GAME STATE> The other player moved an object on their grid. What is your next command?	
6	[A(GM)] MOVE: I, (5,6)		
7		[GM(B)] SAY: I think our grids are aligned now, finished?	
8	[A(GM)] SAY: finished!		

Figure 2: Example episode from Clean Up game. The task is to negotiate a common goal configuration for a number of objects randomly placed on each player's grid, and move them accordingly. Finally, both players have to agree the goal is reached to end the game.

Game Instances All instances are created programmatically by creating different grids. All grids have the same size (7×7), and contain obstacles in form of horizontal and vertical lines, branches, crossings, and corners. They are generated using an Answer Set Programming (ASP) encoding to

ensure all lines are continuous. The maximum number of rounds is fixed to $4 \times n_{obj}$ (where n_{obj} is the object count). Players can collectively accumulate $2 \times n_{obj} + 2$ penalties. We create different sets of experiments by controlling the difficulty along two dimensions: number of empty cells and number of objects. We have three difficult levels. Of the 49 total cells, 34 or 69.4% of cells are empty on the *easy* level, 29 or 59.2% on *medium*, and 24 or 49.0% on *hard*. For each level, we sample three grids, and then place 3, 5, or 7 objects on them, making for **27 instances in total**.

#	Player A	Game Master	Player B
1		[A(GM)] <GAME DESCRIPTION> ... -Items and weights: Ball(2), Book(3), Magazine(1), Rope(4), Lamp(5) -Max Weight: 7 -Your Preferences: Ball=1, Book=2, Magazine=2, Rope=2, Lamp=1	
2	[A(GM)] STRATEGIC REASONING: Ball, Book, Rope have the value-to-weight ratios while respecting the weight limit.		
3	PROPOSAL: Magazine, Ball, Book ARGUMENT: These items balance weight and value well for me. What do you think?	[GM(B)] <GAME DESCRIPTION> ... <Items, weights, weight limit> -Your Preferences: Ball=1, Book=1, Magazine=2, Rope=4, Lamp=5 <PLAYER A Message without STRATEGIC REASONING>	
4		[GM(B)] STRATEGIC REASONING: I can increase the value of the other player's proposal for me by swapping Rope with Book. Let me propose this. PROPOSAL: Rope, Ball, Magazine ARGUMENT: I value Rope highly, but not Book. How about we swap them?	
5		[A(GM)] <PLAYER B Message without STRATEGIC REASONING>	
6	[A(GM)] STRATEGIC REASONING: I value Rope and Book equally. Weight of proposal is $4+2+1=7$ which matches the limit. I should accept. AGREE: Rope, Ball, Magazine ARGUMENT: I am indifferent w.r.t. Rope and Book. Fair deal.		

Figure 3: An example episode of the *Air Balloon Survival* game. Two players must negotiate and argue for their preferred set of items. To reach the goal state, one player must explicitly agree to a proposal made by the other. //Tim-Kristina: Can you add a Strategic Reasoning text (short) to the episode here? - anything just to show this.//

3.3 Air Balloon Survival

This negotiation game evaluates advanced reasoning and interactive collaboration between players, which was previously introduced by Howes et al. (2021) in a different form to study how patients with schizophrenia verbalise their reasoning during social encounters. We have adjusted the task as a two-player game where these players are on a sinking hot air balloon and they have to agree on which

items to keep (or throw out) so that the weight of the balloon is reduced in order to keep floating. Each player has hidden preference values for items and must negotiate to maximize their combined utility score while respecting weight constraints and following structured communication rules. An example episode is given in Figure 3. The game tests individual reasoning through constraint-based optimization (similar to the 0/1 Knapsack Problem) requiring arithmetic and combinatorial search. It also examines collective reasoning through practical rationality and theory of mind, as players must infer their counterpart’s hidden preferences from their responses to reach an optimal agreement. The game concludes when one player explicitly accepts a proposal from the other, making it a concrete test of both computational problem-solving and social negotiation skills. We provide all prompts and other details in Appendix D.

Game Mechanics Both players receive their preference values assigned to items. They are instructed to use specified formats when making proposals and negotiating, and to follow rules such as only being allowed to accept proposals made by the other player. There is no limit on the number of turns, but a game is aborted if no progress is made for eight consecutive turns. Each player may commit up to two mistakes, either by violating game rules or by violating the required syntax. Different from other two games, the players are instructed to also output their STRATEGIC REASONING along with the expected message. Only the expected message is passed to the other player, the strategic reasoning output is used for analysis only. It allows players to “think out loud” about their choices, which can be compared with the reasoning model traces, but we made it an explicit part of the game.

Instance Generation We draw either 15 or 35 items (depending on the experiment) from a randomly generated list constructed by concatenating a capital letter with a two-digit number (e.g., A42, C07). This keeps the naming of items language agnostic and abstract as much as possible from conventional examples of the 0/1 Knapsack Problem models may have seen during training//Tim-Kristina: add the idea behind this design choice - why not go with the names as in Fig 3, why letters?, not to have it depend on any language maybe?...// We assign a weight value to each item drawn from some discrete uniform distribution. The air balloon’s capacity is defined as a fraction

of the combined weight of all items. We experiment with two negotiation levels where we specifically set valuations over items to be the same for players giving them common goals or we invert their preference orderings giving them opposing goals. //Tim-Kristina: what is controlled by negotiation Easy/Hard experiments, write about it, briefly//. Easy level includes the common goal instances //Tim-Kristina: finish the sentence// and the hard level includes opposing goals instances//Tim-Kristina: finish the sentence//. We also experiment with turning the STRATEGIC REASONING off. Lastly, we run complexity experiments by increasing the number of items, using uncorrelated weights and values for the easy case, and correlated ones for the hard case where greedy value-to-weight heuristics for the 0/1 Knapsack fail.//Tim-Kristina: easy means with and hard means that...//. In total, we have six experiments with each having six instances, which leads to the **total number of 36 instances** for this game.

4 Experimental Setup

4.1 Game Instances

Deal or No Deal, *Clean Up* and *Air Balloon Survival* games include 40, 27, and 36 experimental instances, respectively. Each instance is then initialized with defined prompt templates. The same game instances are used for English, German, and Italian experiments because the instances are language-agnostic, only prompt templates have to be aligned for a specific language. All prompt templates are provided in Appendix B.1, C.1, D.1.

4.2 Evaluation Metrics

The *clembench framework* (Chalamalasetti et al., 2023) requires each implemented game to provide two main metrics: % Played and Quality Score. The % Played stands for the percentage of episodes where the evaluated language models followed instructions and the game was terminated by the defined end states. In cases where the game play does not fit the defined states of the game, Game Master either tolerates such behaviour for one or two turns ask the players to try again or simply aborts the game. Game violations can be syntactic (where the game formatting rules are breached, e.g. not including the prefix “SAY” or “MOVE” in the Clean Up game) or semantic (e.g. by attempting to move an object that does not exist in the Clean Up game, or even trying to move another object

into an already occupied cell). The next metric that is required by each game is the *Quality Score*. It is calculated by an objective function to measure how close the played episode came to the target goal. In each game, this metric is calculated for all episodes that have been played, which reached the end state. Once two metrics are calculated, they are aggregated to a single number, the *clemscore* as the normalized product of % *Played* and *Quality Score* (scaled to the interval [0, 100]).

Next, we describe the *Quality Score* details for each game below.

Deal or No Deal For the cooperative game mode, it is the ratio between the achieved total score to the maximum total score for that game instance,

$$\text{Quality}_{\text{coop}} = \frac{\text{Score}_A + \text{Score}_B}{\max(\text{Score}'_A + \text{Score}'_B)}$$

where Score_A and Score_B represent the actual scores achieved by players A and B respectively, and $\max(\text{Score}'_A + \text{Score}'_B)$ denotes the maximum possible combined score achievable for the given game instance.

For the semi-competitive setting, a metric based on Pareto efficiency has been used.

$$\text{Quality}_{\text{semi}} = 1 - \frac{\text{Maximum Pareto Improvement}}{\max(\text{Score}'_i)}$$

where *Maximum Pareto Improvement* represents the largest possible improvement in one player's score without decreasing the other player's score from the current agreement, and $\max(\text{Score}'_i)$ denotes the maximum possible score achievable by any single player i in the given game instance.

Clean Up The *Quality Score* is calculated by multiplying *Distance Score* with *Penalty Score*.

$$\text{Quality} = \text{Distance Score} \cdot \text{Penalty Score} \cdot 100$$

Distance Score ($\text{DS} \in [0, 1]$) is calculated from three components, each representing the sum of Euclidean distances for all identical objects on both grids: the Initial Distance Sum $\mathbf{I}$ at the start of the game, the Final Distance Sum $\mathbf{F}$ at the end of the game, and the Expected Distance Sum $\mathbb{E}$ that approximates the distances for randomly placed objects (see Appendix C.3 for details). The Expected Distance Score ES and Distance Reduction Score RS , quantifying how close the players came

to the goal of perfect alignment of all objects, are calculated as follows:

$$ES = \max \left\{ 0, 1 - \frac{\mathbf{F}}{\mathbb{E}} \right\}; RS = \max \left\{ 0, 1 - \frac{\mathbf{F}}{\mathbf{I}} \right\}$$

DS is then either 0 if final placement is worse than random, or calculated as the mean of both scores:

$$\text{DS} = \begin{cases} \frac{ES+RS}{2} & \text{if } ES > 0 \\ 0 & \text{otherwise} \end{cases}$$

Penalty Score ($\text{PS} \in [0.5, 1]$) is calculated from the penalty count P normalized against max. penalties P_m as follows:

$$\text{PS} = \frac{P_m}{P - 2P_m} + 1.5$$

We chose the hyperbolic function because it is lenient for low P and harsher for P close to P_m .

Air Balloon Survival Each player $p \in P$ receives an individual score based on the utility of the final deal D , i.e. the sum of the preference values of items in D . This sum gets normalized by the value of the optimal solution to their instance of the 0/1 Knapsack Problem. The overall game score is defined as the harmonic mean of the two players normalized scores, further normalized by the maximum harmonic mean achievable for an instance. If the total weight of the deal exceeds the capacity W , the score is set to 0. We chose the harmonic mean as our main metric as it rewards collectively balanced outcomes by favouring deals where both players achieve high scores and penalizing those where one player benefits disproportionately. The individual scores of players is each players outcome relative to the optimal solution of their own knapsack problem. For a final deal $D \subseteq \{1, \dots, n\}$, the normalized score of player $p \in P$ is

$$f_p(D) = 100 \cdot \frac{\sum_{i \in D} v_p(i)}{\text{OPT}_p},$$

where OPT_p is the value of the optimal solution to player p 's 0/1 Knapsack Problem. The harmonic mean of the players' normalized scores is defined as

$$f_{\text{harm}}(D) = \begin{cases} \frac{2f_1(D)f_2(D)}{f_1(D)+f_2(D)} & \text{if } \sum_{i \in D} w_i \leq W, \\ 0 & \text{otherwise.} \end{cases}$$

		GPT-5		GPT-5 mini		CL-4		LM-70B		Nem-9B		Qwen-3		GPT OSS	DS-v3.1
		On	Off	On	Off	On	Off	On	Off	On	Off	On	Off	On	Off
EN	DoND	87.9	32.3	75.7	23.7	94.4	90.1	24.3	43.5	18.5	4.0	57.9	22.5	50.0	59.0
	Cleanup	59.9	42.8	63.4	38.1	69.7	60.0	1.5	19.8	21.8	27.9	55.3	18.6	67.7	57.5
	Air Balloon	98.0	80.1	97.5	82.2	83.8	26.2	2.3	40.9	14.3	13.2	88.7	0	78.2	81.7
	Average	81.9	51.7	78.9	48.0	82.6	58.8	9.4	34.7	18.2	15.0	67.3	13.7	65.3	66.1
	Margin	+30.2		+30.9		+23.8		-25.3		+3.2		+53.6		N/A	N/A
DE	DoND	86.0	19.6	82.9	24.4	80.3	77.1	21.8	53.4	0	4.5	62.1	12.8	34.9	42.4
	Cleanup	78.7	41.2	76.8	41.2	82.7	55.8	3.9	15.4	21.5	0	58.2	14.5	50.1	29.4
	Air Balloon	98.0	81.4	94.9	86.2	85.4	27.8	51.5	43.7	17.4	16.3	76.0	8.8	90.7	75.3
	Average	87.6	47.4	84.9	50.6	82.8	53.6	25.7	37.5	13.0	6.9	65.4	12.0	58.6	49.0
	Margin	+40.2		+34.3		+29.2		-11.8		+6.1		+53.4		N/A	N/A
IT	DoND	78.0	33.8	77.1	26.8	84.4	67.8	17.3	14.3	13.9	11.2	55.6	19.4	29.0	28.9
	Cleanup	89.0	68.8	93.6	77.2	91.6	82.6	6.5	31.1	21.5	12.7	82.6	31.5	63.6	67.6
	Air Balloon	97.2	88.0	97.3	87.8	79.4	27.0	2.7	40.8	0	0	77.8	12.9	83.7	76.4
	Average	88.1	63.5	89.3	63.9	85.1	59.1	8.8	28.7	11.8	8.0	72.0	21.3	58.8	57.6
	Margin	+24.6		+25.4		+26.0		-19.9		+3.8		+50.7		N/A	N/A
Overall Average		85.9	54.2	84.4	54.2	83.5	57.2	14.6	33.7	14.3	10.0	68.2	15.7	60.9	57.6
Overall Margin		+31.7		+30.2		+26.3		-19.1		+4.3		+52.5		N/A	N/A

Table 1: *clemscore* values for three negotiation games on selected LLMs for English, German, and Italian versions. *On*: reasoning mode is turned on, *Off*: reasoning mode is turned off. The best result for each language in each row is highlighted in bold. *CL*: Claude, *LM*: Llama-3.3, *DS*: Deepseek, *Nem*: Nemotron-v2, *DoND*: Deal or No Deal

The final game score is the normalized harmonic mean,

$$\text{Quality}(D) = 100 \cdot \frac{f_{\text{harm}}(D)}{\text{OPT}^*},$$

where OPT^* denotes the maximum harmonic mean achievable in the given instance.

4.3 Evaluated Models

We have evaluated both commercial and open-weight models that have reasoning functionality. We selected *GPT-5*, *GPT-5-mini*¹, *Claude-4* from commercial models². From open-weight models, we selected *Llama3.3-70B* and its reasoning counterpart *Deepseek-R1-distilled-llama-70B*³, *Nemotron-Nano-9B-v2*⁴, *Qwen-3-80B* (*instruction* and *thinking* versions for reasoning off and on modes, respectively), *GPT-OSS-120B* with reasoning on mode only, and finally *Deepseek-v3.1* with reasoning off mode only. For commercial models, we ran them via their respective API backends and we used *OpenRouter.ai* for open-weight models.

¹Reasoning mode for OpenAI models is controlled by setting the *reasoning_effort* parameter to *medium* and it was turned off by setting it to *minimal*

²Reasoning mode is controlled by setting the *thinking* parameter to true/false and providing an upper limit budget (we set it as 1024 tokens)

³Llama3.3-70B model was finetuned to make it a reasoning model by distilling from Deepseek-R1

⁴It is a reasoning model and reasoning can be turned off by including *no_think* in the *system message* and by default it is turned on

5 Results

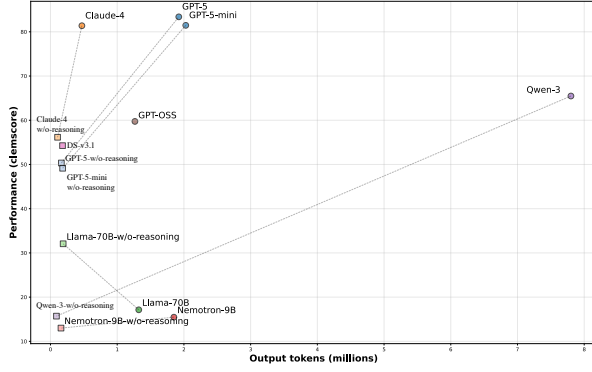
5.1 Overall Performance

We present the results obtained by the models evaluated for English, German, and Italian versions of the negotiation game in Table 1. The models were run with both *reasoning on* and *off* modes.

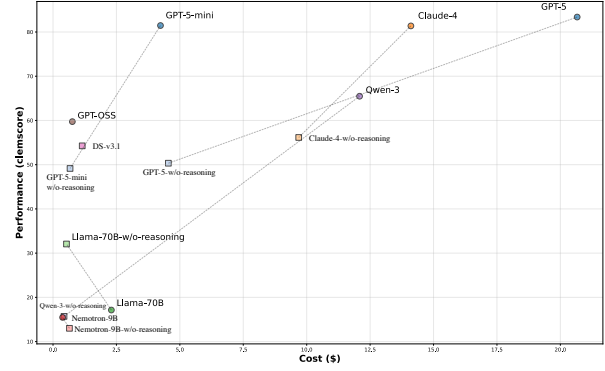
The Effect of Reasoning The most striking observation here is that reasoning mode dramatically improves performance across many models and languages. This is a strong indication that deliberate reasoning significantly enhances strategic game-playing abilities. Only with *Llama-70B* we see different results, which may be the effect of distillation from another model (Deepseek-R1).

Multilingual Capabilities Italian shows the strongest overall performance across most models, with several models achieving scores above 85 points with reasoning enabled. German and English performances are strong too, which suggests these models have particularly good multilingual reasoning capabilities for strategic tasks.

Model Comparison GPT-5 is the clear winner with GPT-5-mini and Claude-4 getting very close performance across all languages. Interestingly, GPT-5 mini sometimes matches or exceeds full GPT-5 performance (particularly in Italian). It terms of performance gap between reasoning on/off modes, Qwen-3 gets the biggest performance jump.



(a) Performance and the number of output tokens for all evaluated models with their reasoning on/off modes averaged for three languages.



(b) Performance and the cost of for all evaluated models with their reasoning on/off modes averaged for three languages.

Figure 4: Trade-off between performance and cost comparison across different models. Results for all languages are averaged.

5.2 Detailed Analysis

RQ1: What is the computational and performance trade-off between reasoning overhead and negotiation effectiveness across different tasks and languages?

Now, we look into at what cost does the performance gap comes in. In Figure 4 a, we present the plot that shows the average performance across all games and languages and how many output tokens it requires. Here, all reasoning and completion tokens are summed up. As expected, the models with reasoning output more tokens than models without it, but Qwen-3 generates almost 7x more tokens than its non-reasoning counterpart. In Figure 4 b, we present the overall cost of experiments for each model. As expected, models with reasoning mode cost more, with *GPT-5* being the most expensive. In terms of deciding on the best trade-off between performance and cost, ***GPT-5-mini* is the most cost-efficient commercial model, and *GPT-OSS* is the best open-weight model** with fraction of the cost compared to Qwen-3.

RQ2: To what extent do models maintain language consistency in their reasoning processes when performing multilingual negotiation tasks?

We ran a language detection script to analyse the reasoning message of the models. The results are presented in Figure 5⁵. We clearly see that all open-weight models mostly generate the reasoning tokens in English whereas Claude-4 “thinks” in the respective language of the task, consistent across all three of them. In Figure 6, we provide the

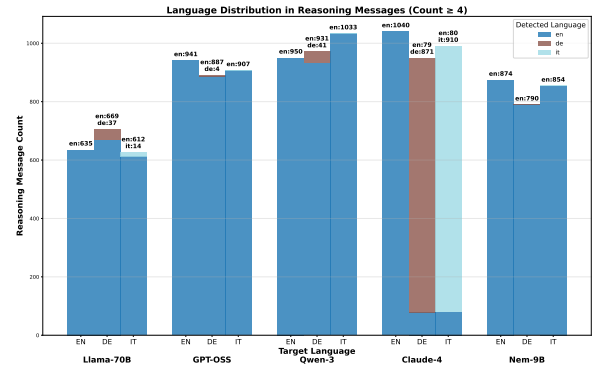


Figure 5: Language distribution in reasoning messages of evaluated models.

heat map of language consistency (whether the language of the message matches the target language) across *completion* and *reasoning* tokens. We can clearly see that *Claude-4* (only commercial model in the analysis) has consistency in both completion and reasoning tokens. Open-weight models retain consistency in completion tokens (thus, abiding the formatting rules as each game requires prefixes in the respective language), but fail to do so with reasoning tokens for German and Italian versions of the games. Thus, we can conclude that **models do not main language consistency in their reasoning processes across multilingual tasks.**

RQ3: Do models demonstrate genuine strategic adaptation over multiple turns, or merely surface-level pattern matching that creates an illusion of learning?

5.3 Findings

Deal or no deal:

Clean Up:

⁵Note: *GPT-5* models do not return reasoning tokens, thus they are excluded from this analysis

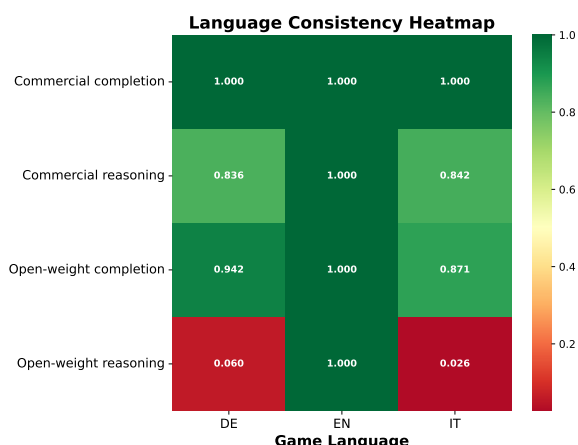


Figure 6: Language consistency across models types (open-weight and commercial) for both completion and reasoning tokens

Air Balloon Survival:

5.4 Qualitative Analysis

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6 Conclusion

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iaculis ante venenatis. Pellentesque mattis sodales eros, a ullamcorper mauris bibendum vel. Fusce iaculis risus quis enim dignissim ornare. Curabitur posuere urna enim, at rutrum libero sollicitudin quis. Nam feugiat, tellus ut vestibulum convallis, ante purus ultrices diam, a accumsan magna magna vel ante.

Limitations

The first limitation is about the scope of negotiation tasks. A large body of work and various other dialogue games exist that we did not include in this study. Another limitation is regarding the analysis of three European languages, which are also high-resource languages. The primary reason for considering only the defined dialogue games is that including more games would result in a higher number of model results, thereby increasing the overall cost of the study. Another limitation is the exclusion of more models from both commercial and open-source ones. As explained above, such decisions would exceed the budget for such a study.

Ethical Considerations

Using paid proprietary APIs with underlying models about which little is known (training data, model architecture) in academic research is less than ideal. Currently, the models tested here support reasoning modes, either on or off (except for GPT-OSS-120B). We hope that open models will include more controls over reasoning aspects and catch up soon in terms of general performance. Deploying such language agents in society has considerable risks. As mentioned in the Related Work, such agents exhibit asymmetric behaviours in negotiations and may expose users to unfaithful actions.

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A Additional Analysis

A.1 Token Usage & Cost

The average token usage and cost for English, German and Italian experiments are given in Figure 7, 8, 9, respectively.

B Deal or No Deal - Game Details

B.1 Prompt Templates

The prompt template for both players of the Deal or No Deal is given in Figure 10.

B.2 Experimental Setup

B.3 Overall Results

B.4 Detailed Analysis

B.5 Qualitative Samples

C Clean Up - Game Details

C.1 Prompt Templates

The prompt templates for both players of Clean Up are given in Figure 11.

Intermittent messages sent by the Game Master to the players are given in Figure 12. Following a player’s response, they are either penalized and reprompted (cf. Figure 12a, 12b), or the turn is passed to the other player (Figure 12c).

C.2 Experimental Setup

Example grids for the three difficulty levels are given in Figure 13.

C.3 Scoring

The expected distance of two objects on an $w \times h$ grid is calculated as follows:

- Assume two independent variables i and j on one discrete dimension $\{1, 2, \dots, w\}$
- The expected absolute distance between i and j can be calculated as

$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \sum_{i=1}^w \sum_{j=1}^w |i - j|$$

- The double sum evaluates to

$$\mathbb{E}[|i - j|] = \frac{1}{w^2} \cdot \frac{(w - 1)(w + 1)}{3} = \frac{w^2 - 1}{3w}$$

For two dimensions, we can then calculate the Euclidean distance of objects o_1 and o_2 as follows:

$$\mathbb{E}[|o_1, o_2|] = \sqrt{\left(\frac{w^2 - 1}{3 \times w}\right)^2 + \left(\frac{h^2 - 1}{3 \times h}\right)^2}$$

For a 7×7 grid, we thus get an expected distance of ≈ 4.19 for two randomly placed objects. To calculate the Expected Distance Sum, we simply multiply it by the number of objects.

This is only an approximation, since we neither take into account the actual grid configuration, nor that two objects cannot have the same location on the same grid.

C.4 Overall Results

C.4.1 Overall Results across levels

As explained in section ??, we designed three broad game difficulty levels based on the ratio of empty cells in a grid. We hypothesize that the denser the background grid is, the more likely the target locations clash with occupied cells, and therefore the more likely models get penalized in scoring. In addition, at each difficulty level, there are three sub-levels: either 3, 5, or 7 objects are placed on

		Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking
Optimal	Explicit maximization	0	1	0	3	11	3	11	0	1	0	0	0	5
	Simple negotiation	17	16	0	7	3	2	6	2	5	5	0	0	8
	Luck	0	0	0	0	0	0	0	0	0	0	1	0	0
Suboptimal	Unclear preferences	1	1	1	1	0	1	0	0	1	1	0	0	0
	Unclaimed items	0	0	0	0	0	0	0	0	0	1	0	0	1
	Claiming valueless item	1	2	0	0	0	2	0	1	3	4	0	0	0
	Wrong proposal	0	0	1	0	0	0	0	0	1	0	0	0	0
Failed	Premature proposal	0	0	3	1	3	0	1	0	3	0	3	1	0
	Wrong proposal	0	0	3	7	1	11	0	13	1	5	0	0	4
	Premature proposal	1	0	12	1	1	0	1	2	4	0	14	19	0
Aborted	Wrong syntax	0	0	0	0	0	1	0	1	1	4	0	0	1
	Non existant items	0	0	0	0	0	0	0	0	0	0	2	0	0
	No proposal	0	0	0	0	0	0	0	0	0	0	0	0	1

Table 2: Game outcomes for the cooperative DoND game mode.

		Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking
Optimal	Explicit maximization	0	1	0	1	0	0	0	0	0	0	0	0	0
	Simple negotiation	18	13	1	6	14	3	9	1	6	5	0	0	6
	Luck	0	0	2	0	0	0	0	0	2	0	2	0	0
Suboptimal	Unclear preferences	0	0	0	1	0	0	0	0	0	0	0	0	0
	Unclaimed items	0	0	0	0	1	1	2	5	0	1	0	0	3
	Claiming valueless item	1	3	0	5	2	1	2	3	0	2	0	0	2
	Wrong proposal	0	0	0	0	0	0	0	0	0	0	0	0	0
Failed	Premature proposal	0	0	1	0	2	0	0	0	0	0	2	1	0
	Wrong proposal	0	3	5	6	1	12	6	8	7	2	0	0	2
	Premature proposal	1	0	7	0	0	1	1	1	4	1	14	18	6
Aborted	Wrong syntax	0	0	4	1	0	0	0	2	1	9	2	0	1
	Non existant items	0	0	0	0	0	0	0	0	0	0	1	1	0
	No proposal	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 3: Game outcomes for the semi-competitive DoND game mode.

	Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking	Instruct
Tell exact values	0	3	1	13	19	19	17	12	14	1	0	0	13	17
Tell only relative values	20	17	0	5	0	1	0	7	0	18	0	0	2	1
Do not tell preferences	0	0	7	2	0	0	2	0	5	1	0	0	5	0
Do not communicate at all	0	0	12	0	0	0	0	0	1	0	20	20	0	2

Table 4: Model willingness to share its value function for the cooperative DoND game mode.

	Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking	Instruct
Tell exact values	0	1	0	11	0	4	0	9	5	0	0	0	6	6
Tell only relative values	19	19	1	9	15	14	8	11	7	20	0	0	1	10
Do not tell preferences	1	0	8	0	5	0	11	0	7	0	0	0	12	0
Do not communicate at all	0	0	11	0	0	0	1	0	1	0	20	20	1	4

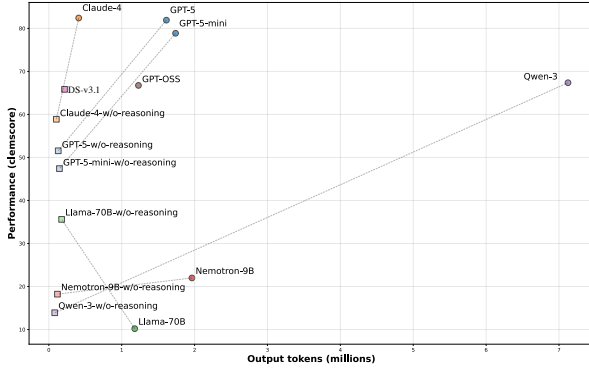
Table 5: Model willingness to share its value function for the semi-competitive DoND game mode.

	Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking	Instruct
Lie about preferences	1	1	0	9	0	0	0	1	0	2	0	0	0	0
Dont lie about preferences	19	19	20	11	19	20	19	18	20	18	20	20	20	20

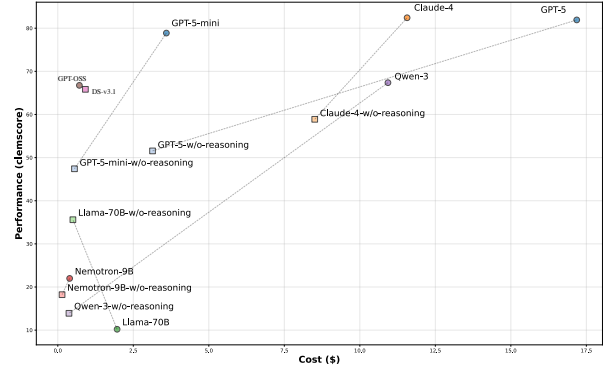
Table 6: Model truthfulness about its value function for the cooperative DoND game mode.

	Claude Reason	No reason	Deepseek r1-distilled	v3	GPT5 Reason	No reason	GPT5-mini Reason	No reason	GPT-OSS	LLAMA	Nemotron Reason	No reason	Qwen3 Thinking	Instruct
Lie about preferences	0	5	0	6	0	0	0	2	1	3	0	0	0	1
Dont lie about preferences	20	15	20	14	20	18	20	18	19	17	20	20	20	0

Table 7: Model truthfulness about its value function for the semi-competitive DoND game mode.

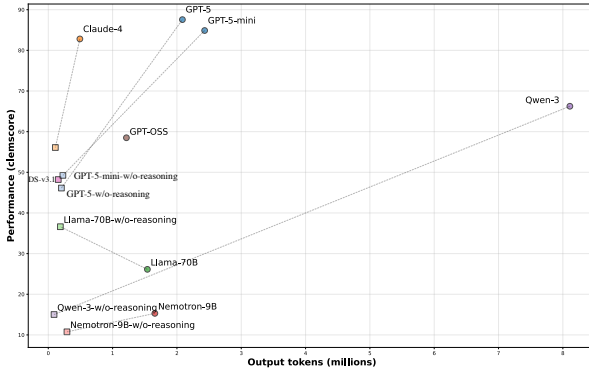


(a) Performance (clmscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for English.

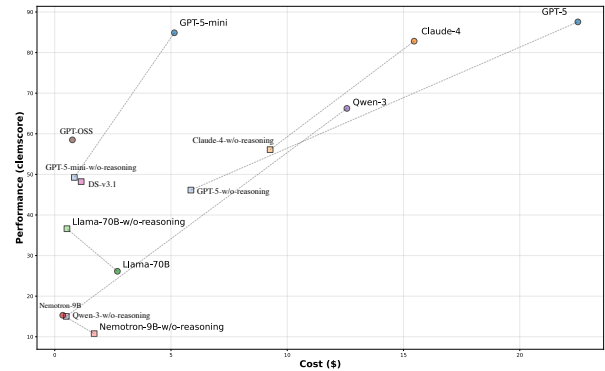


(b) Performance (clmscore) and cost for all evaluated models with their reasoning on/off modes for German.

Figure 7: Trade-off between performance and cost comparison across different models. Results for English.



(a) Performance (clmscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for German.



(b) Performance (clmscore) and average number of output tokens for all evaluated models with their reasoning on/off modes for German.

Figure 8: Trade-off between performance and cost comparison across different models. Results for German.

the grid, on the assumption that the more objects there are, the longer the gameplay traces are, and the less likely that models achieve a high score with mere luck.

Figure 14 plots the main score; clear stratifications of performance across the number of objects are observed in easy and hard levels. When we break down the Main Score into Distance Score and Penalty Score in Figure 15, the stratification at the medium level can also be found.

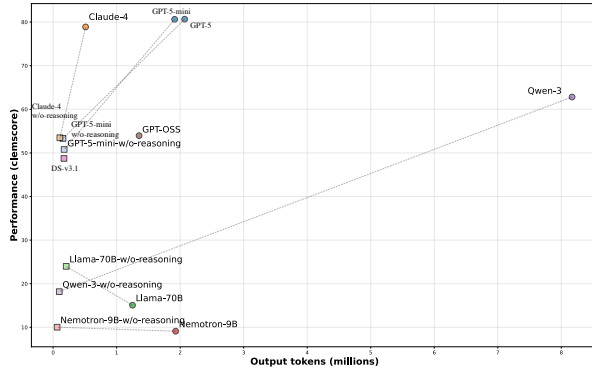
Games with 3 objects achieve the highest scores, regardless of background empty cell ratios, which corroborates our assumption that games with a low number of objects are easier. However, levels with 5 objects have the lowest Distance Score in the medium level and the lowest Penalty Score in both easy and hard levels, indicating that there are factors other than the number of objects that affect performance.

C.4.2 Overall Results across model properties

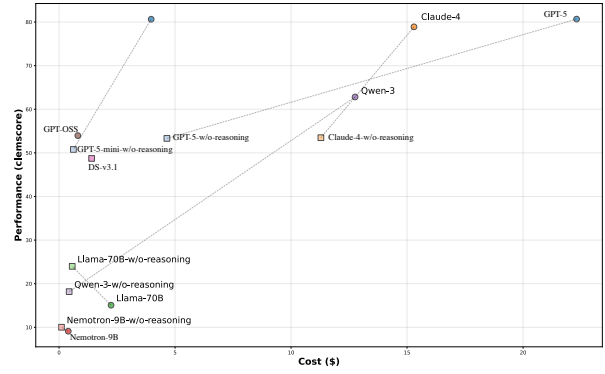
Figure 16 illustrates that the Main Scores of commercial models are skewed toward the upper end of the scale, with a clear concentration in the 75100 range, particularly under reasoning mode. In contrast, open models generally score lower, with many results in the 050 range. Notably, the dots are much sparser for open models when reasoning is on. This is because the Main Score is not recorded when the maximum number of penalties is reached; in these cases, the game will be marked as aborted.

In Figure 17, we further investigate the distribution of Distance Score and Penalty Score. The aborted games, while not shown in the Main Score plot, are represented here with an "x" marker. We set the Penalty Score to 0.5 when the models reach the maximum allowed penalties. If they incur one more penalty, the game is marked as aborted, and the Main Score won't be recorded.

In the Penalty Score subplot, all game plays with



(a) Performance (clemscore) and cost for all evaluated models with their reasoning on/off modes for English.



(b) Performance (clemscore) and cost for all evaluated models with their reasoning on/off modes for Italian.

Figure 9: Trade-off between performance and cost comparison across different models. Results for Italian.

Penalty Score < 0.4 are marked with "x" by definition. While the total number of game plays in each of the four lanes is the same, we see that the open reasoning models have the highest share of aborted games, followed by open non-reasoning models. In the Distance Score subplot, we see that the aborted games span widely for open non-reasoning models, indicating some game plays, although achieving a high Distance Score, have pitifully accumulated too many penalties and thus have been marked aborted and taken out of the account of the Main Score. In contrast, commercial models have generally better Penalty Score and Distance Score.

An interesting observation is that, in the Distance Score subplot, the aborted games tend to reside in lower ranges for non-reasoning models, while the reasoning models might play an aborted game despite achieving a high Distance Score. This is especially prominent for open non-reasoning models, and hints that the reasoning mode, rather than helping in games where these models might achieve a high Distance Score, might introduce noises that lead to more penalties.

C.5 Detailed Analysis

Idea: We hypothesize that Instruction Following ability and Spatial Estimation abilities serve as the foundational pillars of performance, as illustrated in Figure 18. With weaker pillars, model reasoning might not add much to the performance.

Penalties include the Invalid Format Penalty and the Invalid Move Penalty. We plan to use the ratio of Invalid Format Penalty as an agent of Instruction Following ability, sort the models by it, compare the result of the same model with reasoning on or off, and see what percentage the reasoning capabil-

ity adds to the performance. The assumption is that models with better Instruction Following abilities benefit more from the introduction of the reasoning mode.

We plan to apply the same analysis to Invalid Move Penalty.

C.6 Qualitative Samples

Idea: We plan to do qualitative analysis from the following four angles: Spatial Estimations, Strategy, Critical Thinking, Communication & Collaboration, and analyze and compare performance across commercial models (Y/N) and reasoning models (Y/N). For GPT-5, we compare across model size (original / mini) as well.

Here we briefly motivate each of these angles. Concrete examples and analysis will be added later.

C.6.1 Spatial Estimation

From this angle, we check if models can propose valid moving plans. It captures models' Spatial Estimation ability in our grid settings and serves as a foundational pillar of the overall performance. When models are not capable in this regard, they tend to propose invalid moving plans, and attempts to resolve the wrong movements after getting penalties often conduce to cascades of invalid moving plans, therefore quickly exhaust the number of allowed penalties and end up either aborting the game, or attain a low penalty score.

C.6.2 Strategy

In addition to the crude Spatial Estimation, we check how models come up with moving plans. We have observed some smart strategies and their not-so-smart counterparts. In some game plays, one model uses the initial coordinates of the ob-

TEMPLATE B.1.1

You are given three images, one is called target and the other two are distractors. Your task is to generate a referring expression that best describes the target image while distinguishing it from the two other distractor images. The first image is <IMAGE_POSITION>, the second image is <IMAGE_POSITION>, and the third image is <IMAGE_POSITION>.

Instruction: Describe the target image. Generate the referring expression starting with the tag "Expression: " for the given target image. Omit any other text.

Target image: <IMAGE_PATH>
Second image: <IMAGE_PATH>
Third image: <IMAGE_PATH>

TEMPLATE B.1.2

Expression: \$EXPRESSION\$

TEMPLATE B.1.3

You are given three images. You are also given a referring expression that describes one of the given images. Your task is to select the image that matches the given referring expression. Generate only the number (in text) of the image that matches the given expression by selecting first, second, or third.

TARGET_EXPRESSION

Question: Which image does the expression refer to? Start with the tag "Answer: ", followed by your selection. Omit any other text.

First image: <IMAGE_PATH>
Second image: <IMAGE_PATH>
Third image: <IMAGE_PATH>

TEMPLATE B.1.4

Answer: \$ANSWER\$

(a) Prompt template for Player A (Instruction Giver) in the Reference Game.

(b) Prompt template for Player B (Instruction Follower) in the Reference Game

Figure 10: Reference game prompt templates for both players

jects to instruct the other model as target locations, thus avoiding penalties of moving objects to cells occupied by the background grids to a large extent; while in some other game plays, models appear to exchange initial coordinates just for the sake of exchanging, without making use of this information. Another example is in some game plays, models choose the most sparse row as the target, and lay objects one by one on it; while in some other game plays, the models unnecessarily chose geometric patterns to lay the objects, such as on the diagonal, disregarding the occupied cells, and run into complexities that they could have dodged.

C.6.3 Critical Thinking

Some models show the ability to critically reason about the moving plan that either itself or the other player proposed. They either refute the plan immediately when it's proposed, or take a sharp brake when they are just about to execute the invalid moving plan. This is, as expected, more frequent in reasoning models compared to non-reasoning models. However, on some occasions, we see pseudo-reasoning: a model rejects a correct plan, utters reasoning-like sentences, and proposes another plan that sometimes is correct, sometimes is

wrong.

C.6.4 Communication & Collaboration

The above three dimensions positively correlate with the model performance: The better the models are at them, the higher the scores. This dimension, however, is different. Specifically, we observe how models communicate with each other to recover from invalid moves due to target locations being occupied. Some models go at lengths to execute their alternative plans without letting their partner know, they might still have a good score, simply because both player 1 and player 2 adopt the exact same strategy, for example, move the object to one spot next to the original target location. This leads to inflated scores that can be debunked when two models of different default recovery strategies, or one model and one human, play against each other.

D Air Balloon Survival - Game Details

D.1 Prompt Templates

The prompt template for both players of the Hot Air Balloon Survival game is given in Figure 19. Parse error prompts handed to players when their responses did not follow the format instructions

TEMPLATE C.1.1

I am your game master, and you are playing a collaborative game with the following grid as game board:

1234567

C	o	o	o				1
o	I	o					2
o	o						3
o	o						4
							5
o				o	W	T	6
o				o	H		7

* The upper edge displays x-coordinates increasing to the right, and the right edge y-coordinates increasing downward.

* The following objects are randomly placed on your grid: 'W', 'I', 'T', 'C', 'H'.

The other player sees a variation of the game board, where the objects are placed at different random locations. You cannot see the other player's board, and they cannot see yours.

Goal:

Both players need to move the objects on their respective background so that identical objects end up at the same coordinates. You have to communicate with the other player to agree upon a common goal configuration.

Rules:

* In each turn, you can send exactly one of the following two commands:

1. ``SAY: <MESSAGE>``: to send a message (everything up until the next line break) to the other player. I will forward it to your partner.

2. ``MOVE: <OBJECT>, (<X>, <Y>)``: to move an object to a new position, where ``<X>`` is the column and ``<Y>`` is the row. I will inform you if your move was successful or not.

* If you don't stick to the format, or send several commands at once, I have to penalize you.

* If both players accumulate more than 12 penalties, you both lose the game.

* It is vital that you communicate with the other player regarding your goal state! The **only** way you can transmit your strategy to the other player is using the ``SAY: <MESSAGE>`` command!

Moving Objects

* You can only move objects to cells within the bounds of the grid. The target cell must be empty, i.e., it must only contain the symbol 'o'.

* If you try to move an object to a spot that is not empty, or try to move it outside of the grid, I have to penalize you. You get another try.

* Before making a move, double check that the target spot is empty, and does not hold any letter, frame, or line!

End of Game

If you think you reached the goal of aligning all objects, you can ask the other player to finish the game by sending ``SAY: finished?``. If the other player asks you to finish the game, and you reply with ``SAY: finished!``, the game will end.

Both players win if the game ends within 20 rounds, where one round is defined as two players each sending a valid command.

Scoring:

The closer the identical objects are in both game boards, the more points you get. Penalties reduce your points. Can you beat the record?

TEMPLATE C.1.2

Please send a message to the other player to start the game!

TEMPLATE C.1.3

The other player started the game by sending this message:

`"<START_MESSAGE>"`

What is your first command?

(a) Prompt template for Player A in the game Clean Up.

(b) Prompt template for Player B in the game Clean Up.

Figure 11: Clean Up prompt templates for both players

TEMPLATE C.1.4

Penalty: <REASON>

Make sure that your response only contains either SAY: <MESSAGE> or MOVE: <OBJECT>, (<X>, <Y>), and nothing else!

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Your message must not contain anything before the command!
- Your message must not contain anything after the command!
- Your message must not contain anything before or after the command!
- Your message contains more than one command!
- Your message is not in the expected format!
- You must begin the game by sending a message to the other player!

(a) Penalty prompt template for format errors in the game Clean Up

TEMPLATE C.1.5

<REASON>

You have collectively accumulated <N> of <M> penalties. Please try again!

<REASON> can be one of the following:

- Invalid move: (<X>,<Y>) is out of bounds.
- Penalty: (<X>,<Y>) is not empty, but contains '<OBJECT>'.
- Invalid move: Your image has no object with ID '<OBJECT>'.

(b) Prompt templates for invalid moves in the game Clean Up

TEMPLATE C.1.6

<LAST MOVE>

You are currently playing round <R> of maximum <MR>. You have collectively accumulated <N> of <M> penalties. <OTHER PLAYER ACTION>
What is your next command?

<LAST MOVE> can be:

- Your message has been relayed to the other player.
- Moved '<OBJECT>' to (<X>,<Y>) successfully. Your updated grid looks like this:
<GRID>

<OTHER PLAYER ACTION> can be:

- The other player sent this message:
"<MESSAGE>"
- The other player moved an object on their grid.

(c) New turn prompts in the game Clean Up

Figure 12: Intermittent prompts in the game Clean Up

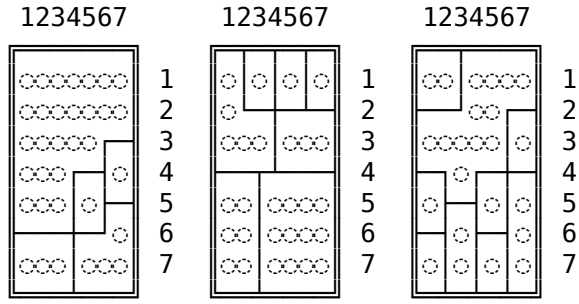


Figure 13: Example grids for *easy*, *medium*, and *hard* levels in Clean Up.

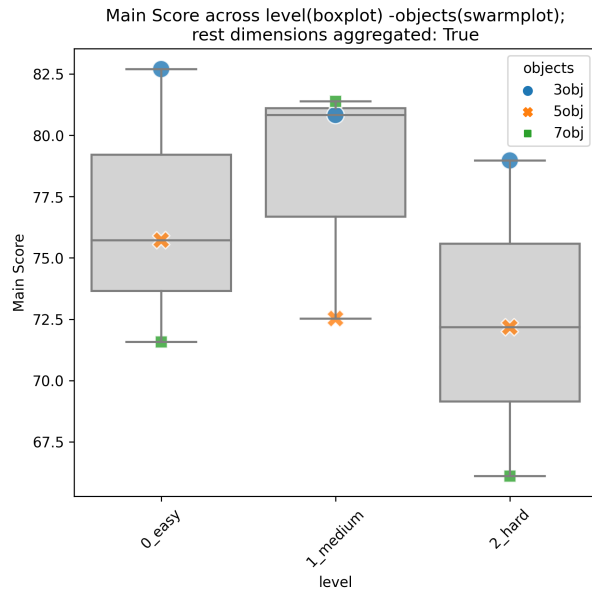


Figure 14: Main Score across levels and sub-levels

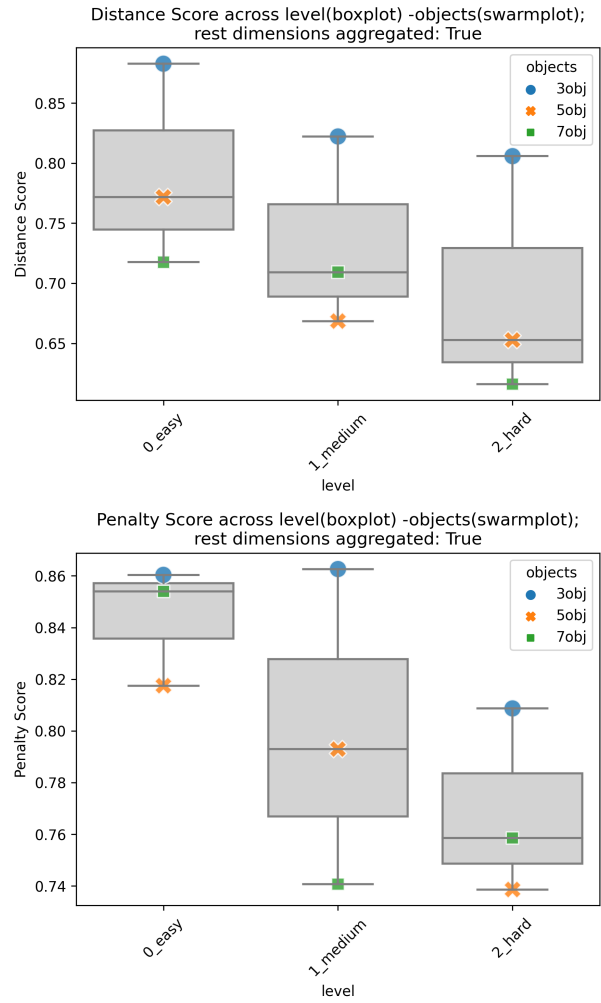


Figure 15: Sub Scores across levels and sub-levels

are given in Figure 20, and game error prompts for invalid actions are given in Figure 21.

Substrings Figure 19 marked by \$. . . \$ are placeholders which get replaced depending on the game instance. Note that the last three placeholders may also be replaced by the empty string depending on the game settings. Here we list the replacements for placeholders in the initial prompt:

- \$LIMIT\$ maximum weight for a deal.
- \$ITEM_WEIGHTS\$ mapping from items to weights.
- \$UTILITY_SCALE_PLAYER\$ mapping from items to preference values for a given player.
- \$STRATEGIC_REASONING_FORMAT\$

STRATEGIC REASONING: {'...'}
Describe your strategic reasoning or
anticipation explaining your choice

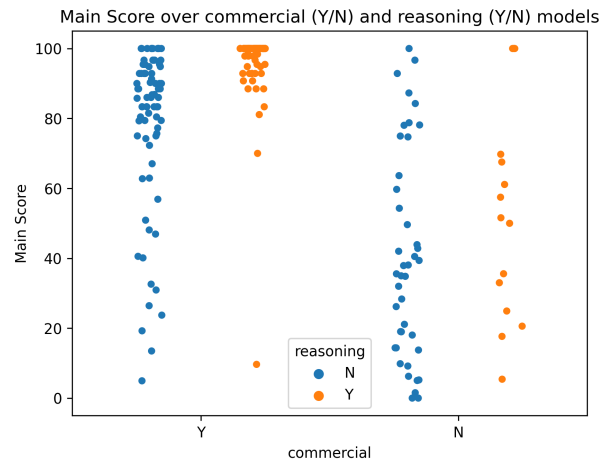


Figure 16: Main Score over different model properties

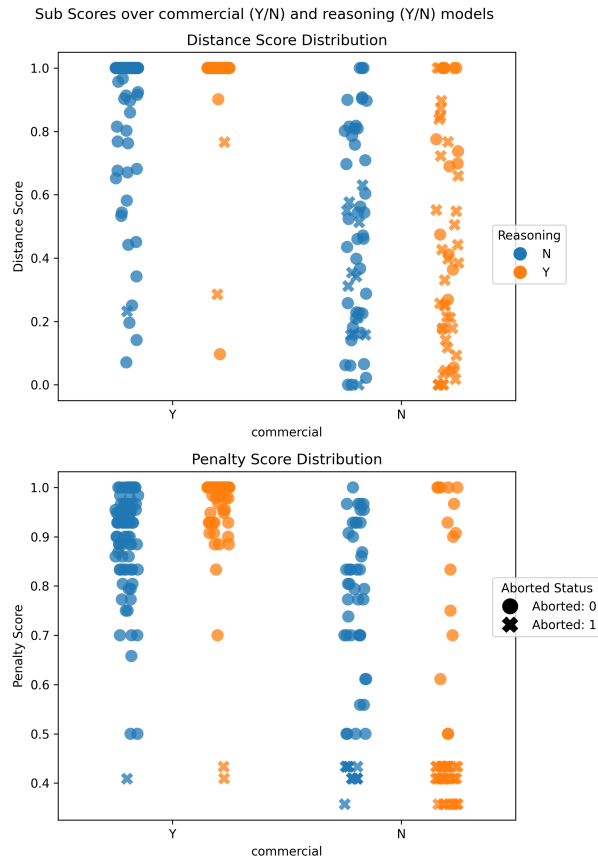


Figure 17: Sub Scores over different model properties

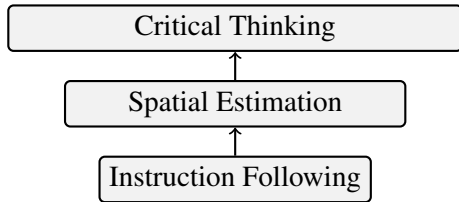


Figure 18: Capability Hierarchical

of action. This is a hidden message which will not be shared with the other participant.

- **\$REQUIRE_ARGUMENT\$** You must include the ARGUMENT format at least once somewhere in all of your messages.

- **\$STRATEGIC_REASONING_RULE\$**

You must include the STRATEGIC REASONING format only once at the very beginning of every one of your messages and not more often. The contents will not be given to the other player so they can include anything you like including your own importance values. Here you should reason

Model	Pareto Adherence
GPT-OSS 120B (reasoning)	0.763
GPT-5 (reasoning)	0.675
GPT-5 Mini (reasoning)	0.548
Qwen3-Next-80B (reasoning)	0.464
Nemotron-Nano 9B v2 (reasoning)	0.367
GPT-5 Mini	0.269
GPT-5	0.217
Claude Sonnet 4 (reasoning)	0.164
DeepSeek Chat v3.1	0.150
Claude Sonnet 4	0.011
LLaMA-3.3-70B Instruct	0.000
Nemotron-Nano 9B v2	0.000
DeepSeek R1-Distill LLaMA-70B	-
Qwen3-Next-80B	-

Table 8: Average per instance Pareto adherence rate for proposals per model. Values were computed on English instances with 15 items.

Model	Alternation
DeepSeek R1-Distill LLaMA-70B (reasoning)	1.000
GPT-5 Mini (reasoning)	1.000
GPT-5	1.000
Claude Sonnet 4	1.000
GPT-5 (reasoning)	1.000
GPT-5 Mini	1.000
GPT-OSS 120B	0.991
Claude Sonnet 4 (reasoning)	0.947
Qwen3-Next-80B (reasoning)	0.823
Nemotron-Nano 9B v2	0.875
DeepSeek Chat v3.1	0.734
LLaMA-3.3-70B Instruct	0.573
Nemotron-Nano 9B v2 (reasoning)	0.167
Qwen3-Next-80B	-

Table 9: Average alternation rate per model on all English instances.

step by step to come up with your next move.

D.2 Experimental Setup

D.3 Overall Results

D.4 Detailed Analysis

D.4.1 Negotiation Dynamics

D.4.2 Collaboration

D.4.3 Critical Thinking

D.5 Qualitative Samples

TEMPLATE D.1.1

You are participating in a collaborative negotiation game.

Together with another participant, you must agree on a single set of items that will be kept. Each of you has your own view of how much each item matters to you (importance). You do not know how the other participant values the items. Additionally, you are given the effort each item demands. You may only agree on a set if the total effort of the selected items does not exceed a shared limit:

LIMIT = \$LIMIT\$

Here are the individual item effort values:

Item effort = \$ITEM_WEIGHTS\$

Here is your personal view on the importance of each item:

Item importance values = \$UTILITY_SCALE_PLAYER\$

Goal:

Your goal is to negotiate a shared set of items that benefits you as much as possible (i.e., maximizes total importance to YOU), while staying within the LIMIT. You are not required to make a PROPOSAL in every message - you can simply negotiate as well. All tactics are allowed!

Interaction Protocol:

You may only use the following structured formats in a message:

PROPOSAL: {'A', 'B', 'C', }

Propose keeping exactly those items.

REFUSE: {'A', 'B', 'C', }

Explicitly reject opponent's proposal.

ARGUMENT: {'...'}

Defend your last proposal or argue against the player's proposal.

AGREE: {'A', 'B', 'C', }

Accept the opponent's proposal which ends the game.

\$STRATEGIC_REASONING_FORMAT\$

Rules:

You may only AGREE on a proposal the other party has logged via PROPOSAL.

You may only REFUSE a proposal the other party has logged via PROPOSAL.

Total effort of any PROPOSAL or AGREE set must be $\leq$ LIMIT.

Do NOT reveal your hidden importance scores.

A tag in a structured format must be followed by colon and whitespace. The argument must be a python set containing 0 or more strings.

So, it must be of the form TAG: {...}

Strictly follow the interaction protocol and DO NOT write anything beyond the given structure.

The game ends when one side gives an AGREE to a PROPOSAL made by the other player.

The content in your response which can be handed to the other player has to be non-empty.

Only proposals which have been logged via the PROPOSAL format structure and which haven't been refused via REFUSE are active.

\$REQUIRE_ARGUMENT\$

\$STRATEGIC_REASONING_RULE\$

Figure 19: Initial prompt template of Hot Air Balloon. Note that for the second player we append newline and the phrase 'You will now receive the first message of the other player.' at the very end of the initial prompt.

TEMPLATE D.1.2

Your response did not start with the proper strategic reasoning tag at the very beginning of your response.
The very first structured format must be of the form STRATEGIC REASONING: {...}. Try again.

(a) Parse error prompt in case strategic reasoning tag at the beginning of a response is missing. Only applies when strategic reasoning tag is required

TEMPLATE D.1.3

Your response did not contain an argument.
You must include the structured format ARGUMENT: {...} somewhere in your response. Try again.

(b) Parse error prompt in case an argument is missing. Only applies when argument tag is required

TEMPLATE D.1.4

Your response contained an untagged sequence or you used STRATEGIC REASONING more than once.
You may only use the structured formats as explained in the initial message.
They must all be of the form TAG: {...}.

(c) Parse error prompt in case the response contains an untagged sequence.

TEMPLATE D.1.5

Your response only contained a strategic reasoning tag.
You must at least include one more valid tag in your response, so that the other player receives a message. Try again.

(d) Parse error prompt in case response only contained a strategic reasoning tag. Only applies when argument tag is not required.

TEMPLATE D.1.6

You used a PROPOSAL tag, but did not provide a valid python set containing strings as arguments, e.g. {'A', 'B', 'C', ...}. Try again.

(e) Parse error prompt in case of an invalid python set as argument.

Figure 20: Parse error prompts handed to player when a response did not follow the instructions on structured formats.

TEMPLATE D.1.7

You refused a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(a) Game error prompt in case a non-active was refused.

TEMPLATE D.1.8

You made more than one agreement.
Final deals cannot be ambiguous. Try again.

(b) Game error prompt in case agreement to a deal was ambiguous.

TEMPLATE D.1.9

You agreed to a proposal which is not active.
Proposals are only active if they have been
logged by the other player via PROPOSAL and
have not been deactivated by you via REFUSE.
Try again.

(c) Game error prompt in case .

TEMPLATE D.1.10

Your proposal includes items which are not in
the game. Try again.

(d) Game error prompt in case a non-active proposal was agreed to.

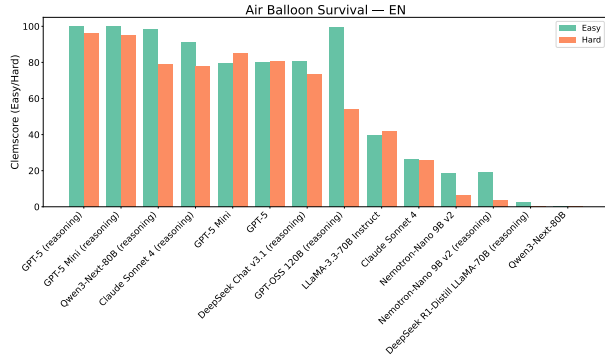
Figure 21: Game error prompts handed to player when a response did not follow the instructions on the rules of the game.

Model	P1	P2
GPT-OSS 120B	96.314	74.077
Nemotron-Nano 9B v2 (reasoning)	92.475	64.105
GPT-5 (reasoning)	89.634	77.853
GPT-5 Mini (no reasoning)	86.698	57.792
Qwen3-Next-80B (reasoning)	85.928	75.960
GPT-5 Mini (reasoning)	85.891	79.753
GPT-5 (no reasoning)	85.010	63.461
DeepSeek Chat v3.1	79.987	56.153
Claude Sonnet 4 (no reasoning)	77.342	64.821
Claude Sonnet 4 (reasoning)	70.231	72.229
Nemotron-Nano 9B v2 (no reasoning)	56.113	61.660
LLaMA-3.3-70B Instruct	52.536	40.910
DeepSeek R1-Distill LLaMA-70B	-	-
Qwen3-Next-80B	-	-

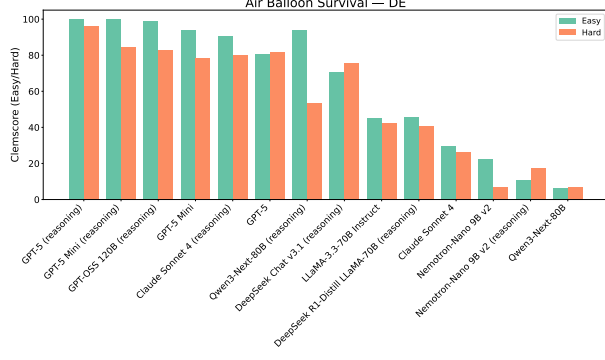
Table 10: Average player scores on English opposing goals instances (only considering played games).

Model	Proactive	Reactive	Total Games
Claude Sonnet 4 (no reasoning)	0	13	13
Claude Sonnet 4 (reasoning)	1	13	14
DeepSeek Chat v3.1	11	12	23
DeepSeek R1-Distill LLaMA-70B	1	0	1
GPT-5 (no reasoning)	15	5	21
GPT-5 (reasoning)	17	3	24
GPT-5 Mini (no reasoning)	19	1	21
GPT-5 Mini (reasoning)	19	5	26
GPT-OSS 120B	5	6	21
LLaMA-3.3-70B Instruct	3	10	19
Nemotron-Nano 9B v2 (no reasoning)	21	1	22
Nemotron-Nano 9B v2 (reasoning)	0	2	2
Qwen3-Next-80B	4	5	15

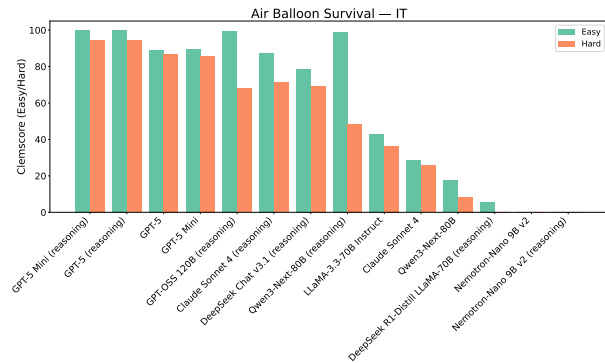
Table 11: Initial reasoning style - classified by counting collaborative (proactive) versus egocentric (reactive) keywords in the first turn of each game within Strategic Reasoning tag. Calculated on all English instances.



(a) CLEMScores on English instances.

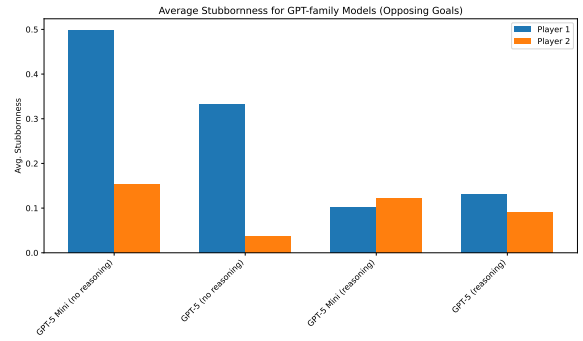


(b) CLEMScores on German instances.

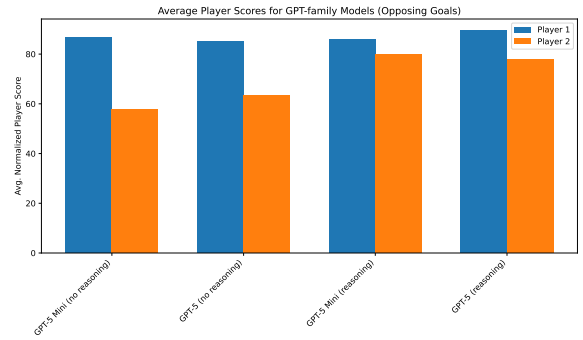


(c) CLEMScores on Italian instances.

Figure 22: Overall model performance for easy and hard sections of experiments for all languages.

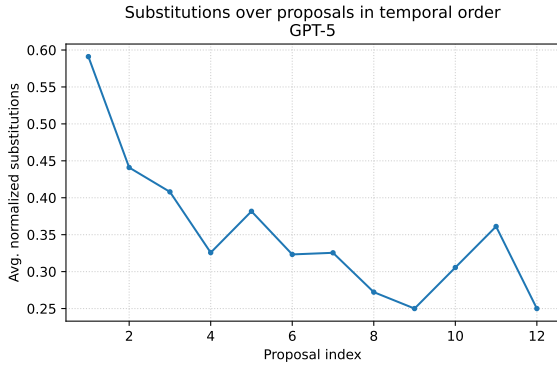


(a) Average stubbornness per player per instance for reasoning vs. conventional GPT-family models.

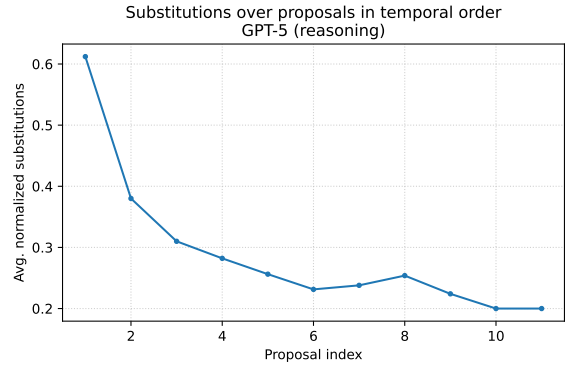


(b) Average player scores for GPT-family models.

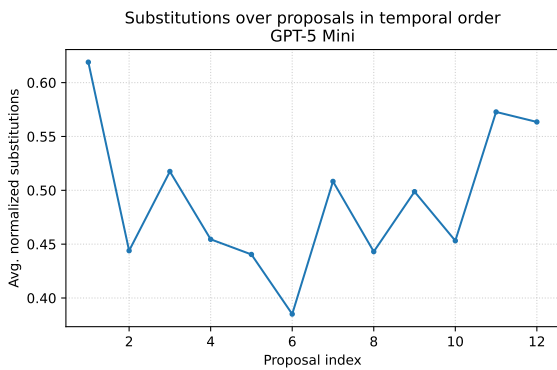
Figure 23: Comparison of player 1 and player 2 in GPT-family models. Values were computed on English instances with opposing-goals setting



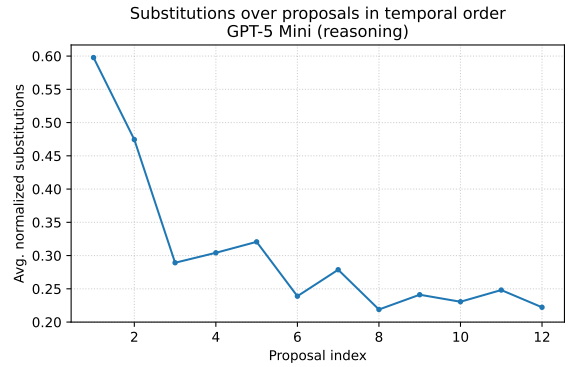
(a) GPT-5 (no reasoning). Curve is decreasing, but less smoothly compared to the reasoning version.



(b) GPT-5 (reasoning). Very smooth and decreasing curve, indicating coherent bargaining and stronger early disagreement.



(c) GPT-5 Mini (no reasoning). Substitutions rise in longer negotiations, reflecting frustration and less coherent bargaining.



(d) GPT-5 Mini (reasoning). Substitutions decrease on average, showing more coherent bargaining than non-reasoning model.

Figure 24: Comparison of bargaining coherence across models. Reasoning in top models amplifies the smoothed negotiation, with each proposal building on the previous. Results are from English experiments, capped at 12 proposals.